Mathematical overview: facts, assumptions and limits

A planar outline can be congruent to its reflected outline under translations and rotations only if it admits an orientation-reversing planar symmetry. The asymmetric F-like polygon on p. 1 has none, so face-up slides/turns cannot match its mirror. The equal-arm L has a diagonal reflection and the T has a vertical reflection, so their mirrors do match face-up. Turning a flat tile over is a proper rotation in three-dimensional space that induces an orientation-reversing map in the tile's plane; all three outline pairs then match. Face dots track permitted moves and are excluded from the matching object. These claims use congruent thin, unwarped tiles and ignore material thickness.

A regular tetrahedron has 12 proper rotations, exactly its even vertex permutations. A red three-edge Hamiltonian path has only two automorphisms on its four vertices: identity and path reversal. Reversal swaps two pairs of vertices and is even. Thus its red/ordinary edge coloring has no improper symmetry, and its mirror is chiral. A three-red star or triangle has an odd symmetry swapping two indistinguishable outer vertices and is achiral. Of the 20 ways to color exactly three of six edges red, 12 are paths (chiral), four stars and four triangles (achiral). Red vertices are **not** the object: all vertices here are uncolored, and temporary letters do not count.

Three mutually perpendicular arms with distinct fixed identities (colors, lengths, or both) form an oriented triad. Proper rotations preserve its handedness and a reflection reverses it. Three equally long arms in three different colors are chiral; two equal colors or all equal colors allow mirror matching. Three different lengths restore chirality even when colors coincide. Repeated colors do not remove chirality from all objects; other features may still distinguish the arms. Claims concern whole rigid models, not flexing, exchanging sleeves or disassembling during a match.

K–1/2–3 can explore dimension change with flat tiles. Grades 2–3 can build/compare corners when handling is reliable. Grades 4–5 can explain chirality in new edge/corner objects and justify all possible turns. No permutation parity or vector determinants are child prerequisites. Unsuccessful attempts are experiments, not impossibility proofs; symmetry or orientation arguments provide certificates.

Materials, entry route and launch

Per pair: six thin outline tiles (F and mirror, equal-arm L and mirror, T and mirror), two congruent rigid regular tetrahedral frames, six removable red edge sleeves and six ordinary sleeves or bare edges, two rigid equal-arm three-way right-angle connectors with three 60 mm arms each, six removable R, two B and two G **arm** sleeves for the equal-arm pair, and a second pair of rigid connectors with arms 30/60/90 mm finished red before class (or six additional red arm sleeves if not permanently colored). Temporary A/B/C/D labels help build; remove or cover them before matching. Use no flexible joint that changes the corner angles. Six pair-sized full kits for K–1 / 2–3 / 4–5 need 36 outline tiles, 12 tetrahedral frames, 36 red **tetrahedron-edge** sleeves (and 36 ordinary sleeves if used), 12 equal-arm corner models with 36 R/12 B/12 G removable **arm** sleeves, and 12 already-red unequal-arm models. If the unequal arms also need removable sleeves, add 36 red arm sleeves. That full stock is optional: practical minimum for the proposed route is two outline kits at K–1, two outline/corner kits at 2–3, and two complete kits at 4–5: 36 outline tiles, four tetrahedral frames with twelve red edge sleeves, eight equal-arm corner models with 24 R/8 B/8 G arm sleeves, and four already-red unequal-arm models. If those unequal arms need sleeves, add twelve red arm sleeves. Reuse base tetrahedral frames and ask an adult to prepare all attachments before class. A rigid three-way orthogonal building connector can supply the corner; choose one that holds rods perpendicular without play, and check each angle against a square before use. Keep an exact model and its mirror available together. A flexible bendable joint or planar Y is not a substitute. Available rigid frame components are preferred; a new equipment purchase is not assumed. This is a preparation suggestion, not a physical test performed here.

Allow handling, then use a non-task rectangular tile: slide and turn it with the face dot up, then lift and turn it over. Show that the dot tracks the allowed move but is ignored in comparing the outline. Give one matching pair to confirm transfer. At the older tables demonstrate putting a sleeve on a whole tetrahedral edge, keeping vertices uncolored. For corners, check all three pairwise angles are right angles and arms stay rigid. Use the student cube-corner → isolated three-arm → whole-model turn visual, then show a true rotated copy before the mirror comparison. The oblique picture projects three

spatial right angles to unequal planar angles; it is not a flat Y to build on.

A manageable first route is P1 for 15–25 minutes, then P3's equal-length corners for 15–30. Reserve P2 and the unequal-length continuation for later return visits, 20–40 minutes each including explanation. These are unpiloted estimates. K–1 needs adult reading and face-up rule enforcement; the parent may hold the fixed comparison while children move their tile. Older children should own model choices and matching attempts. Counts are at most six edges; arithmetic is minimal. Three anchored adults suffice with prepared models. Actual rigidity, angle accuracy, sleeve handling and children's model transfer have **not** been physically pretested; if the exact rigid models cannot be supplied, defer the spatial matching investigation rather than claim the picture performed it.

Solutions, hints and extensions

P1 / student p. 1, flat tiles in two versus three dimensions. In printed top-to-bottom order, face-up answers are no, yes, yes. For L, reflecting in its equal-arm diagonal leaves it fixed; composing that reflection with the mirror construction gives a face-up quarter-turn matching the pair. For T, its own vertical mirror symmetry makes the shown mirror identical up to placement. The F's left upright and unequal horizontal branches force a unique orientation, and reversing its handedness cannot match that orientation. Once turning over is allowed, yes for all three. Digital verification tests exact polygon corner distance matrices and orientation for every cyclic/reversed vertex correspondence; it does not certify cutting tolerances. Hint: align one distinguishing corner/arm, then compare the rest without lifting. Extension: make a new outline whose mirror matches face-up and one whose mirror needs turning over. Test a partner's construction; do not infer chirality just from a letter's name.

P2 / student p. 2, two-color edge chirality. Build with A top, B left front, C right front, D rear; dashed edges run to rear D. Mirrors retain each temporary vertex's identity under geometric reflection. The displayed red edges are AB/BC/CD (path), AB/AC/AD (star), and AB/AC/BC (triangle). Answers are no, yes, yes. To change the path's answer move red CD to AC, forming a triangle. To change the star move AD to CD, making path B-A-C-D. To change the triangle move AC to CD, making path A-B-C-D. Preserve exactly three red edges in each edit and remake its mirror.

For the path, a model match must take one endpoint to an endpoint and then follow the unique red path. This permits only the two vertex assignments identity/reversal; both preserve spatial orientation. A mirror matching would need to reverse orientation, so neither works. With the star, keep the unique degree-three red vertex aligned; the other three vertices are indistinguishable and may be cyclically turned to align. With the triangle, keep its unique omitted vertex aligned and rotate the three indistinguishable triangle vertices. This supplies actual match routes as well as the adult odd-symmetry explanation. Hint: compare red degrees at the four vertices; identify the forced vertex maps before turning. Extension: classify all 20 three-red patterns by degree lists: star (3,1,1,1), triangle (2,2,2,0), path (2,2,1,1). Ask whether the degree list alone certifies the mirror result; here those are all cases, but that is not a general graph theorem.

P3 / student p. 3, rigid corner handedness. Equal-length R/B/G: no mirror match. R/R/G: yes; R/R/R: yes. When all colors are equal, lengths 30/60/90 mm distinguish arms and make a mirror match impossible again. Exact model: common endpoint O=(0,0,0), arms along +x,+y,+z, each length 60 mm. Mirror exchanges x and y. With distinct colors R on x, B on y, G on z, a match would have to exchange those two directions while leaving the third fixed, an orientation reversal. Equal colors supply an indistinguishable-arm exchange; a proper rotation can then match. The unequal-arm version uses endpoints (30,0,0),(0,60,0),(0,0,90) and the same mirror x↔y; its unique arm lengths force the forbidden exchange.

A child certificate: point the directed G arm toward the viewer on both models, using the common joint as the center of the plane perpendicular to G. Turn each whole model around G until R points to the viewer's right. B then points up in one model and down in the mirror. Equivalently, the shorter quarter-turn from R to B around that joint has opposite directions in the two views. A proper rotation preserving the directed G arm preserves this quarter-turn direction; the joint and fixed viewing plane supply the reference that a bare cyclic list of two colors would lack. For length-only models point the longest arm toward the viewer and the shortest to the right; the middle arm is up versus down. Hint: save the aligned comparison so repeated turns do not erase the reference. Extension: mix repeated

colors with unequal lengths, or remove an arm; compare which identities remain distinguishable. Do not automatically transfer the tetrahedral repeated-label theorem to a differently decorated corner.

Distinction and verification

Base Week 37 colors tetrahedral vertices, merges repeated labels, counts 12 rotations and two four-label classes. New P1 changes ambient dimension for planar tiles; P2 colors edges and finds chirality with only two colors; P3 changes the geometry to a right-angle corner and distinguishes arms by length. These are three different objects/questions, not three relabelings of the base tetrahedron. They are original constructions informed by the base overview and Week 1 encore format; no downloaded problem is copied.

student/verify.py checks exact planar outline congruences; all 12 even and 12 odd tetrahedral permutations; all 20 three-red edge patterns; and all 24 proper signed-axis rotations of the corner models. The verifier confirms the printed examples and the listed edit types can be checked by the same action. These finite checks support, rather than replace, the overview's general explanations. student/build.sh OUTPUT_DIR is portable and uses only standard Python plus ReportLab for authoring. Record use separately as W37-B1/B2/B3, with actual cases attempted. Everything remains unpiloted; physical model pretests are unperformed.